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3D Architecture of Porous and Tight Dolomites Using High-Resolution Machine Learning workflow in a Carbonate Reservoir

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Abstract

Dolomite rocks play a crucial role in reservoir characterization and 3D modeling in the studied carbonate reservoir. Their rock properties vary greatly due to factors, such as: formation mechanisms, dominant mineralogy, original fabrics and diagenesis. This study uses a Geologically High-Resolution Machine Learning (GHREM) workflow to investigate the 3D distribution of two main dolomite types - porous/permeable and tight - affecting reservoir quality, heterogeneity, and architecture.

Stage 1: Multi-seismic attributes such as acoustic impedance, relative impedance, root mean square, and quadrature amplitude volumes were used as the input to generate high-resolution bulk density and acoustic impedance properties. Well-based bulk density and acoustic impedance logs served as calibration targets for this stage. Stage 2: The resulting high-resolution 3D properties were then used as inputs for predicting the final target properties (i.e. the porous and tight dolomite 3D discrete model). The final GHREM-based prediction incorporated 1D dolomite flags as discrete 1D input. A neural network algorithm drove the machine learning process in both stages, achieving better predictive accuracy over alternative approaches.

The GHREM workflow utilized an in-house machine learning application which enables an objective characterization of the 3D architectural patterns of the primary dolomite facies between wells (porous and tight dolomites). By integrating multiple seismic attributes, intermediate high-resolution bulk density and acoustic impedance 3D volumes, and 1D dolomite identifier flags, this method provided a robust framework for identifying and quantifying 3D dolomite body geometries. Blind testing validated the efficacy of the employed neural network algorithm, yielding accuracies exceeding 85%. Stratigraphically, distinctive distribution patterns emerged, revealing varying proportions of porous and tight dolomites across different intervals. Notably, tight dolomites predominate within the transgressive systems tract of sequences, particularly in the basal portion of the reservoir, whereas their spatial dominance shifts towards the flanks of structural features. Conversely, porous dolomites primarily occupy the regressive systems tracts of sequences, manifesting as narrow, discontinuous bodies with reduced spatial extent compared to tight dolomites. Ultimately, the resultant 3D porous and tight dolomite models served as critical inputs for constructing the comprehensive Integrated Diagenetic and Depositional Facies (IDDF) model (as outlined in Zeiza et al., 2023), facilitating subsequent generation of corresponding 3D porosity and permeability models.

Dolomitization in the regressive systems tract interval, specifically in association with pre-cursor reefal facies (i.e. *Cladocoropsis* & *Stromatoporoid* grainstones/rudstones) exhibits a strong correlation with the diagenetic process of dissolution. This diagenetic interaction results in the formation of a touching vug network, which significantly contribute to fluid-flow pathways. Petrophysical analysis utilizing whole-core computed tomography (CT) scanning reveals these vuggy dolostones possess elevated porosity and permeability values.

Introduction

Dolomitization is a ubiquitous diagenetic process in carbonate rocks that has varying impacts on reservoir properties, with its timing and influence on porosity and permeability remaining a topic of ongoing discussion and research and is summarized well in publications such as in Warren (2000), Machel (2004) and more recently by Yang *et al* (2020). For the carbonate rocks addressed by this study, understanding the genesis and distribution of the type of dolomite was critical for predicting reservoir quality. Reservoir intervals influenced by the late stage hydrothermal linked dolomitization was found to have a dual effect by either retaining porosity (dolomitization of pre-existing secondary porosity generated dissolution enhanced pathways and to a lesser extent recrystallization within muddy facies), or reducing it (via extensive recrystallization and grain replacement). Lucia and Rahnis (1998) outlined that the extensive amount of large crystalline dolostone found around vugular pore systems in this reservoir suggest that the precursor limestone pore-space was utilized by the dolomitizing fluids that have therefore inherited the architecture of the original rock. This is corroborated by a more recent analysis of borehole image logs, highlighting that heavily dolomitized zones are found around areas of the reservoir which have incurred total losses while drilling, with images exhibiting larger geometries of dissolution being the primary culprit. At a field wide scale (Figure 1), the integration of flow meter results with dolomite distribution data visibly shows that there is an enhancement of flow in zones which exhibit the greatest proportion of discontinuous dolomite. Therefore, the net effect on reservoir performance is often controlled by the spatial distribution of tight versus porous dolomite facies.

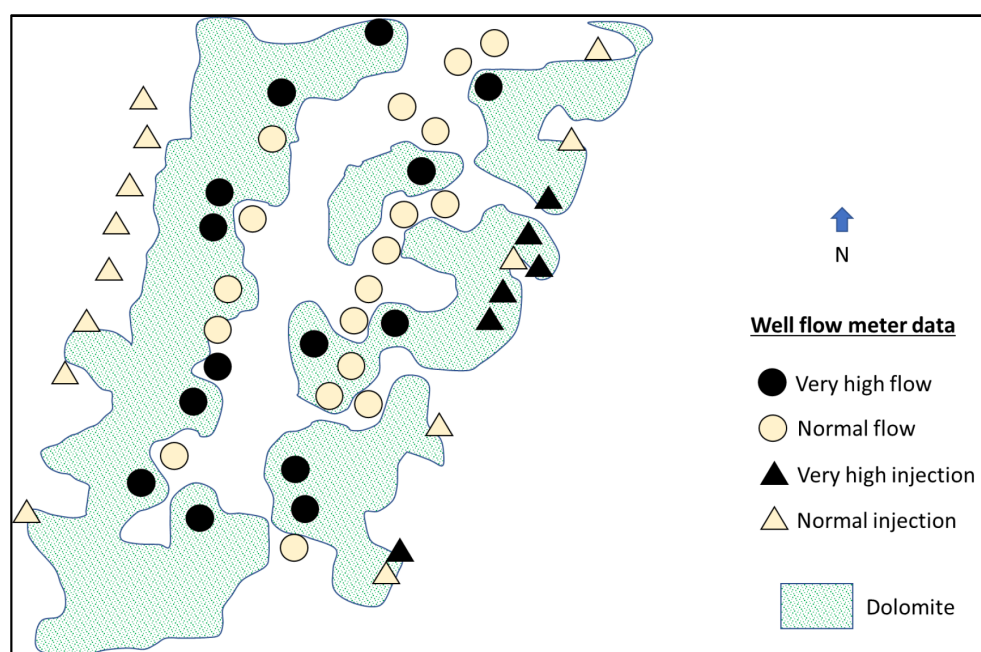


Figure 1—Conceptual cartoon of a portion of the study area showing the distribution of wells with normal as opposed to very high flow rates, that are coincident with zones of greater dolomite distribution. This also holds true for injectors, that demonstrate greater injectivity when they were placed in areas of greater dolomite coverage.

With the increasing utilization of machine learning techniques in geologic interpretations and reservoir property predictions, great promise for enhancing subsurface characterization has been demonstrated by recent studies (Nguyen et al., 2022; Pan et al., 2022; Tomski et al., 2022). However, the potential of machine learning is not without its limitations, as it can be susceptible to errors and overlook critical subsurface features, thereby compromising accuracy (Pyrz, 2022). To mitigate these risks and unlock the full potential of machine learning processes, we recommend three guiding principles: (1) combining robust subsurface data with prior geological knowledge to develop informed models, (2) subjecting machine-generated results to rigorous expert evaluation to ensure accuracy and validity, and (3) ensuring that geoscience-related machine learning projects are led by experienced geoscientists who can provide contextual understanding and refine the outputs. By adhering to these guiding values, we can effectively harness the benefits of machine learning while minimizing its drawbacks, ultimately leading to more reliable and accurate subsurface predictions.

Building on this foundation, our study presents a good example of customizing a high-resolution machine learning workflow to characterize the 3D architecture of porous and tight dolomites in the subsurface. By integrating detailed core characterization, logs and tailored seismic-derived attributes, this approach has demonstrated the spatial variability of diagenetic overprints and their petrophysical expressions, particularly those tied to both porosity-enhancing and destructive dolomitization. The workflow outlined in this paper had the ability to highlight subtle geological patterns often overlooked in conventional workflows, enabling a more predictive and geologically consistent reservoir model. Our results demonstrate how data-driven techniques can discern the stratigraphic and diagenetic controls on reservoir quality. This integrated workflow provides a predictive and geologically consistent framework for modeling the distribution of porous and tight dolomites, ultimately supporting more effective reservoir characterization and development strategies in complex carbonate settings.

Method and Workflow

The GHREM (Geologically High-Resolution Machine Learning) workflow is designed to incorporate geologically consistent, high-resolution 3D property volumes, thereby enhancing the accuracy and reliability of machine learning-based geological predictions (Figure 2). This workflow integrates multidisciplinary datasets—including well logs, one-dimensional discrete indicators of the porous and tight dolomite, multiple seismic attributes, and high-resolution 3D volumes of petrophysical properties such as acoustic impedance and density. By leveraging these integrated data sources, GHREM facilitates the spatial prediction of porous and tight dolomite distributions in three dimensions. Generally, dolomite is characterized by higher acoustic impedance values compared to limestone, owing to its distinct seismic properties (Lucia, 2007). The higher acoustic impedance of dolomite is primarily attributed to its greater seismic velocity and density, which are influenced by its mineralogical composition and diagenetic history. In particular, the acoustic impedance of dolomite is often higher than that of limestone due to the replacement of calcium by magnesium, which increases the density and seismic velocity of the rock (Wang et al., 2017; Xu et al., 2019). Furthermore, the acoustic impedance of dolomite can vary significantly depending on its porosity and fluid saturation (Han et al., 2020), with porous dolomite typically exhibiting lower acoustic impedance values than tight dolomite. Accurate identification of both porous and tight dolomite is essential for enhancing characterization in diagenetically altered carbonate reservoirs, as it directly influences reservoir quality prediction, improves volumetric accuracy, and supports the formulation of more effective field development strategies.

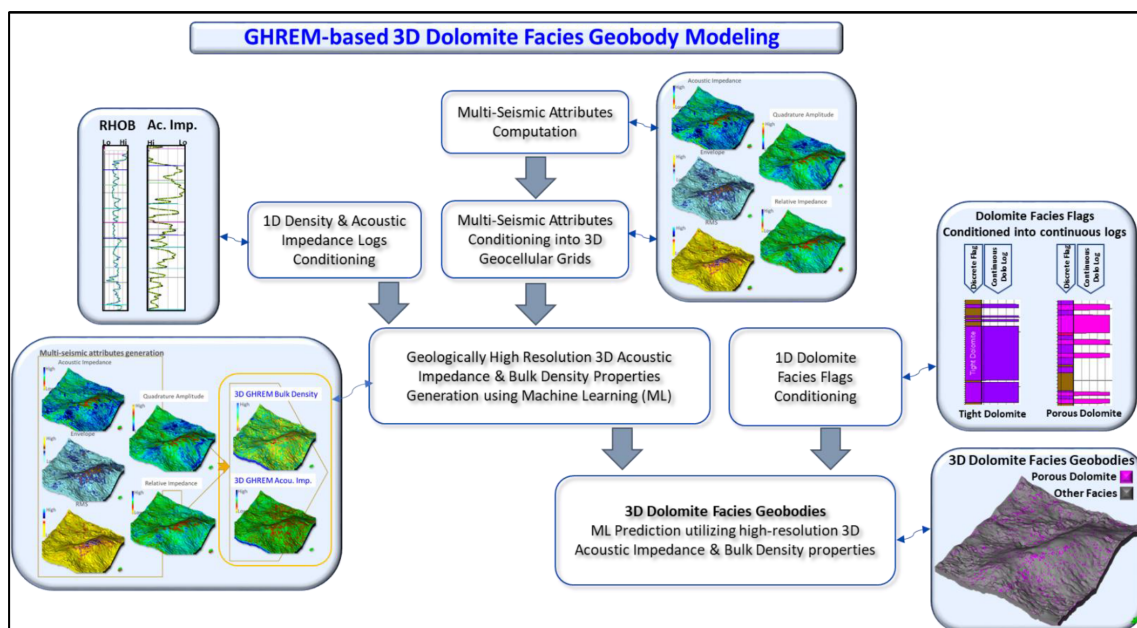


Figure 2—Geologically High-Resolution Machine Learning (GHREM) Workflow for Dolomite Facies 3D Modeling

The GHREM workflow involves two key stages: (1) generation of high-resolution three-dimensional (3D) acoustic impedance and bulk density volumes, using supervised machine learning algorithms, and (2) prediction of 3D distributions of porous and tight dolomite facies bodies through machine learning, incorporating the previously generated acoustic impedance and bulk density volumes as input properties (Figure 3). These machine learning stages leveraged an in-house machine learning platform, which facilitated the development of robust and reliable model realizations through the integration of advanced algorithms and data analytics, ultimately enhancing the accuracy and efficiency of the predictive modeling process.

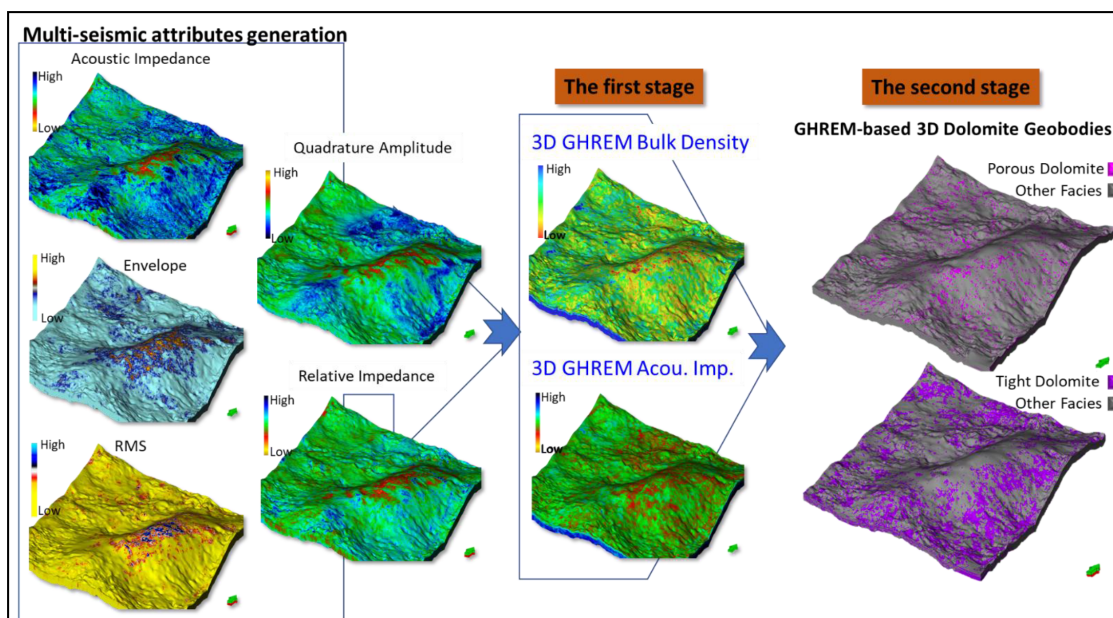


Figure 3—Multi seismic attributes are utilized to generate (1) High-resolution 3D Acoustic Impedance and Bulk Density Properties, and eventually (2) GHREM-based 3D Porous and Tight Dolomite Geobody model

In the initial stage of the GHREM workflow, a targeted suite of seismic attributes best suited to capture the dolomite included—Acoustic Impedance, Quadrature Amplitude, Root Mean Square (RMS) Amplitude, Envelope, and Relative Impedance—are derived from the conventional seismic dataset and incorporated as input features into a machine learning framework (Figure 3). These attributes were selected based on their established geophysical relevance and effectiveness of their observed correlation with dolomitic and non-dolomitic facies distributions, with testing of other attributes ongoing. Their integration enhances the model's ability to predict the spatial distribution, geometry, and petrophysical variability of the porous and tight dolomite facies geobodies, supporting more accurate subsurface characterization in carbonate reservoir systems.

The Quadrature seismic attribute was phase shifted to yield in a higher resolution and better alignment with reservoir properties. In the context of the study area, higher Quadrature amplitude values are generally correlated with increased porosity (Alaudah et al., 2019; Tleel et al., 2020), particularly in porous dolomite intervals and vice versa with tight dolomite. This attribute was found to be a useful input for dolomite prediction and population in the machine learning prediction due to its sensitivities to porosity variation.

Acoustic impedance attributes are derived through seismic inversion, transforming seismic reflection data into quantitative representations of subsurface elastic properties (Russell et al., 2018; Zhang et al., 2020). Specifically, acoustic impedance—defined as the product of rock density and seismic velocity—serves as a critical attribute for delineating heterogeneities. Calibrating seismic attributes with well-log measurements improves the prediction of petrophysical properties, such as tight and porous dolomitic facies. For instance, porous dolomite has low acoustic impedance due to lower bulk density and seismic velocity, whereas tight, show the opposite. This inverse relationship between porosity and acoustic impedance enhances the capacity to identify and predict reservoir quality within carbonate systems, hence, accurate delineation.

Seismic relative impedance (Eq. 1) normalizes seismic attribute enhances interpretation by removing large scale trends and highlighting variations (Russell et al., 2016). In carbonate reservoirs, relative impedance has proven particularly effective for distinguishing lithofacies with subtle petrophysical contrasts. Porous dolomites typically yield lower relative impedance values due to reduced acoustic velocity and density, while tight dolomites exhibit higher relative values, reflecting elevated impedance linked to low porosity and increased compaction. These impedance anomalies provide a quantitative basis for mapping lateral facies variations and identifying potential reservoir zones, even in settings where absolute impedance contrasts may be muted.

$$\text{Relative Impedance} = \frac{Z_{\text{Actual}} - Z_{\text{Reference}}}{Z_{\text{Reference}}} \quad \text{Eq.1}$$

Where:

Z_{actual} is the observed acoustic impedance

$Z_{\text{reference}}$ is a smoothed or modeled background impedance.

The Root Mean Square (RMS) amplitude attribute is derived by computing the square root of the mean of the squared seismic amplitude values within a defined time window (Schultz et al., 2015; Chopra et al., 2017). This attribute effectively highlights lateral variations in seismic reflectivity and serves as a proxy for the energy content of the seismic signal. Elevated RMS amplitude values are frequently associated with porous reservoir intervals, such as the porous dolomite geobodies, where increased porosity and fluid saturation enhance acoustic impedance contrasts and reflection strength and vice versa with tight dolomites.

For strata discontinuities and heterogeneities, the envelope attribute was used to decouple amplitude, enabling the identification of lithologic heterogeneities and strata discontinuities (Wang et al., 2020). These variations are associated with porous dolomite or other diagenetic overprints in such carbonate systems (Wang et al., 2017). Envelope attribute provides critical input for the delineation of subtle subsurface features that may not be readily resolvable using conventional amplitude displays alone, owing to the inherent limitations of single-attribute analysis. By integrating the Envelope attribute with other seismic

attributes, we can improve subsurface predictions and accuracy of geological interpretations in such diagenetically behaving system.

Collectively, the workflow integrates multi-seismic attributes into three-dimensional (3D) geocellular model. This integration involves a geometric resampling procedure to generate mean values from the seismic attributes and assign it to a cell, thereby providing a comprehensive dataset of numerical properties that can be utilized as input for subsequent machine learning analyses. This process enables the creation of a robust and accurate 3D representation of the subsurface, facilitating the utilization of advanced data analytics and machine learning techniques to improve geological interpretations and predictions.

The initial stage of the machine learning workflow employs a comprehensive dataset comprising the previously mentioned seismic attributes, in conjunction with a predefined stratigraphic zone property. The step is intended to estimate high-resolution three-dimensional (3D) Acoustic Impedance properties, via combining 3D seismic attributes and well-based acoustic logs for training. To ensure robustness and accuracy, multiple machine learning iterations were performed using Extreme Gradient Boost (XGB) and Neural Network (NN) algorithms. The resultant high-resolution properties facilitated an accurate prediction of both porous and tight dolomite facies, providing a robust foundation for subsequent analyses and estimations to aid on interpretations and predictions.

Similarly, the workflow was also applied to produce a high-resolution Bulk Density, which complements the Acoustic Impedance property to predict porous and tight dolomite geobodies. A machine learning-based approach is employed to estimate the Geologically High-resolution Bulk Density volume, leveraging the previously mentioned attributes, in conjunction with a stratigraphic zonation. Well logs were utilized as both training data and prediction targets, enabling a robust and accurate estimation of the bulk density property. As an extra step, multiple machine learning estimations were performed using Extreme Gradient Boost (XGB) and Neural Network (NN) algorithms to examine different realizations.

The final step of the GHREM process uses advanced data to predict the location of dolomite formations with high accuracy. To improve the predictions, the data is prepared by resampling it to a finer resolution of 0.2 ft. This data is then used to train machine learning algorithms, including Random Forest, Extreme Gradient Boost, and Neural Network, to classify the formations. After testing, the Neural Network algorithm is found to produce the most accurate and realistic results, creating detailed 3D models of the dolomite formations. These models are consistent with geological principles and can be used to inform decisions about oil and gas exploration and production. The use of Neural Network in this process helps to create a more comprehensive and accurate framework for understanding reservoir properties and behavior.

This workflow combines machine learning with advanced data to predict the complexity of carbonate reservoirs with high accuracy. The result is a powerful tool for understanding and modeling these reservoirs, which can help the oil and gas industry make better decisions about where to drill and how to extract resources. By using GHREM approach to analyze detailed data, it can reduce uncertainty and improve the accuracy of reservoir models, leading to more efficient and effective hydrocarbon recovery. This can have a significant impact on the industry, enabling better planning, drilling, and production strategies.

Results and Discussion

High-resolution acoustic impedance 3D property volumes were estimated through a series of machine learning experiments that leveraged seismic attributes as input features and well-log-derived acoustic impedance as the prediction target. A range of supervised learning algorithms—including Random Forest (RF), Neural Networks (NN), and Extreme Gradient Boosting (XGB)—were systematically evaluated based on their predictive performance and geological consistency. Among the models tested, XGB yielded the most accurate and geologically coherent results, particularly when applied in conjunction with the previously mentioned seismic attributes and stratigraphic zone property, as detailed in [Table 1](#). Similarly,

the same process was applied to generate high-resolution bulk density prediction. Prediction runs values are presented in [Table 2](#).

Table 1—Most optimal machine learning models for 3D High-Resolution Acoustic Impedance property prediction

Run #	Input	Target	Algorithm	Accuracy
1	Acoustic Impedance, Relative Impedance, Quadrature Amplitude, Root Mean Square (RMS), Envelope & Zone Property	Hi-Res Acoustic Impedance	XGB	88%
2		Hi-Res Acoustic Impedance	XGB	87%
3		Hi-Res Acoustic Impedance	XGB	85%
4		Hi-Res Acoustic Impedance	XGB	83%
5		Hi-Res Acoustic Impedance	XGB	81%

Table 2—Most optimal machine learning models for 3D High-Resolution Bulk Density property prediction

Run #	Input	Target	Algorithm	Accuracy
1	Acoustic Impedance, Relative Impedance, Quadrature Amplitude, Root Mean Square (RMS), Envelope & Zone Property	Hi-Res Bulk Density	XGB	88%
2		Hi-Res Bulk Density	XGB	88%
3		Hi-Res Bulk Density	XGB	87%
4		Hi-Res Bulk Density	XGB	83%
5		Hi-Res Bulk Density	XGB	82%

One of the highest-performing machine learning models achieved an 88% cross-validation accuracy ([Fig. 4A](#)), indicative of robust predictive performance. This level of accuracy was enabled by integrating multiple seismic attributes along with a stratigraphic zone parameter, the relative importance of which is illustrated in [Figure 4B](#). The resulting high-resolution three-dimensional (3D) acoustic impedance volume shows strong agreement with the input acoustic impedance well log data ([Figs. 4C and 4D](#)), revealing higher confidence in stratigraphically and structurally coherent geometries and trends ([Fig. 4E](#)) that is consistent with the expected distributions of porous and tight dolomite facies.

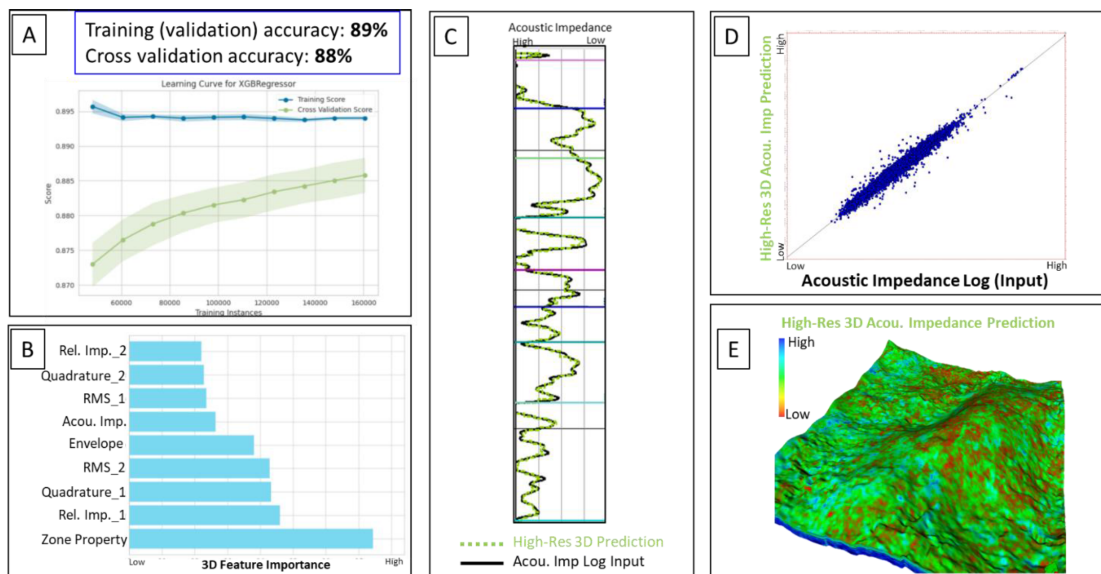


Figure 4—Results of machine learning-based high-resolution acoustic impedance prediction: (a). Model training validation accuracy of 89% and cross-validation accuracy of 88%, indicating robust performance and minimal overfitting; (b). feature engineering of the input results; (c). log view of modeled vs actual acoustic impedance from one of the wells (d). cross plot of modeled vs actual acoustic impedance from several wells; (e). 3D visual of the resultant acoustics impedance volume.

The most optimal machine learning model achieved a cross-validation accuracy of 88% (Fig. 5A) in predicting 3D bulk density, demonstrating strong predictive capability with the help of the seismic attributes and stratigraphic zonation. Figure 5B. The resulting high-resolution 3D bulk density volume shows a strong positive correlation with the input bulk density log data (Figs. 5C and 5D), reinforcing the model's reliability. Moreover, the predicted 3D property exhibits spatially coherent geometries and trends (Fig. 5E) that align with the expected distribution of porous and tight dolomite facies. This congruence between the predicted bulk density and geological expectations validates the model's robustness and highlights its potential as a reliable tool for predicting dolomite-prone facies and characterizing complex carbonate systems.

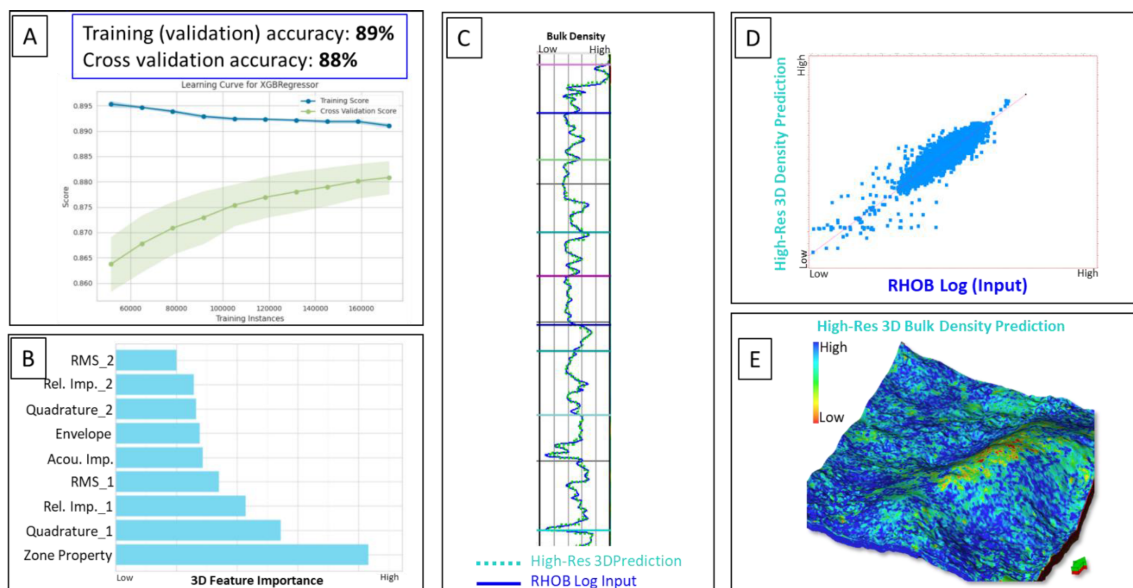


Figure 5—Results of machine learning-based high-resolution bulk density prediction: (a). Model training validation accuracy of 89% and cross-validation accuracy of 88%, indicating robust performance and minimal overfitting; (b). feature engineering of the input results; (c). log view of modeled vs actual bulk density from one of the wells (d). cross plot of modeled vs actual bulk density from several wells; (e). 3D visual of the resultant bulk density volume.

High-resolution 3D acoustic impedance and bulk density volumes were then integrated with stratigraphic zonation to predict dolomite geobodies (Figure 6 and 7). Machine learning model evaluation identified a top-performing configuration with a model accuracy of $\sim 85\%$ (Figure 6A and 7A), indicating strong predictive performance. Feature importance analysis (Figure 6B and 7B) showed a relatively balanced influence among the ten high-resolution input properties, with a slight dominance of several high-resolution attributes—highlighting a complex, multivariate interplay. This integrated approach enabled accurate prediction of porous and tight dolomite facies geobodies, essential for delineating reservoir architecture and informing strategies for enhanced hydrocarbon recovery.

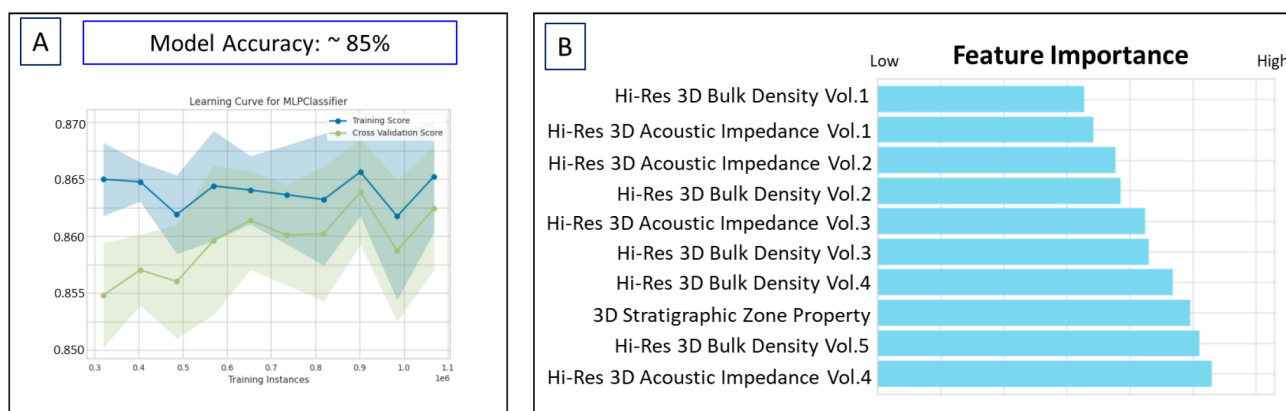


Figure 6—GHREM-based porous dolomite facies prediction results: (a). $\sim 85\%$ model accuracy achieved; (b). 10 high resolution 3D properties (acoustic impedance, density and stratigraphic zone) utilized as input, with varying feature importance.

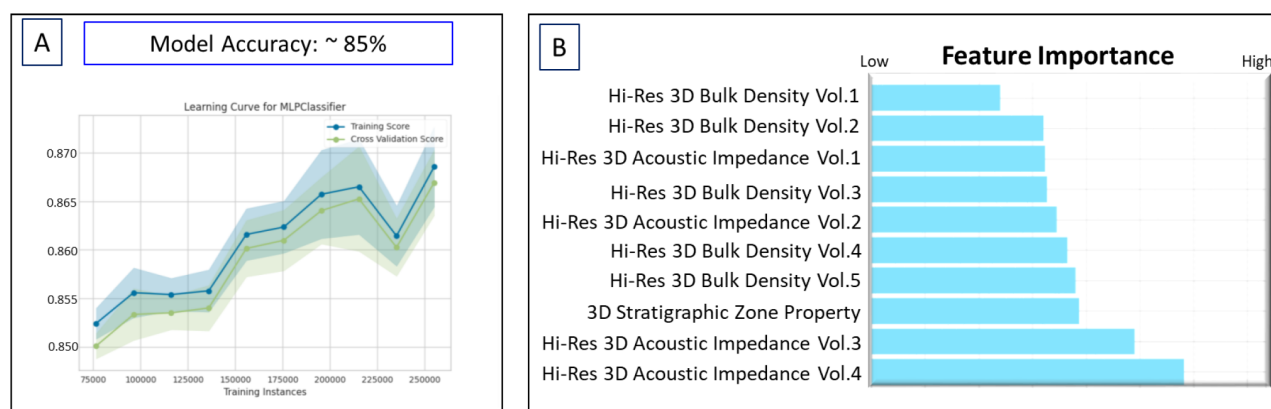


Figure 7—GHREM-based tight dolomite facies prediction results: (a). $\sim 85\%$ model accuracy achieved; (b). 10 high resolution 3D properties (acoustic impedance, bulk density and stratigraphic zone) utilized as input, with varying feature importance.

The GHREM approach demonstrates strong predictive capability as a quantitative and objective method for modeling the three-dimensional distribution of dolomitized carbonate reservoir, i.e. porous and tight dolomite geobodies, as illustrated in Figure 6 and 7, in a machine learning enhanced workflow. The results highlight pronounced contrasts in the 3D architectural distribution between porous and tight dolomite facies, reflecting their distinct nature.

GHREM analysis reveals that porous dolomite is significantly less abundant than tight dolomite across the study area. The results show that porous dolomites are typically confined to thin, laterally discontinuous streaks with limited spatial extent (Figure 8). These features are primarily observed within the upper stratigraphic units, occurring predominantly along the mid-flank to deep-flank structural positions. In contrast to the diagenetically enhanced vuggy porous limestone facies, which are predominantly developed at the structural crest (Zeiza et al., 2025), porous dolomite is restricted to isolated occurrences that

are intermittently found in the crestal areas (Figure 8). The distribution of porous dolomite within the study area is interpreted to be controlled by diagenetic processes linked to pre-cursor depositional facies, dolomitizing fluid flow pathways and localized geochemical conditions. The dolomitizing fluids were likely sourced from the deep-seated fluid flow along high-permeability conduits formed by pre-cursor depositional fabrics. These conditions favored selective replacement of limestone in high-energy grain dominated facies preserving intercrystalline, moldic and vuggy porosity (Saller and Henderson, 1998).

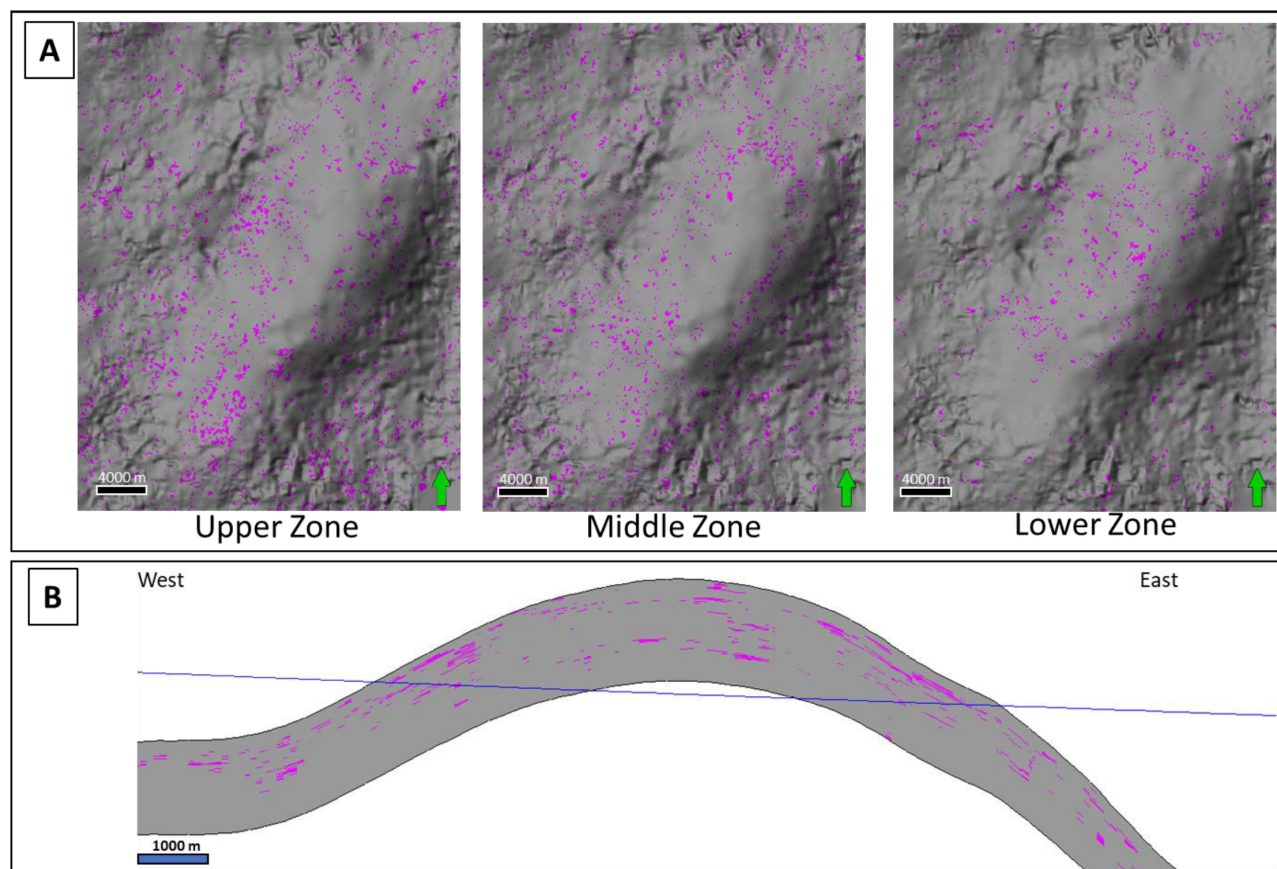


Figure 8—GHREM-based porous dolomite facies distribution results: (a). Map view of porous dolomite facies distribution in the upper, middle and lower zone; (b). Cross section view of the porous dolomite facies distribution.

The distribution of porous dolomite facies is found to be strongly tied to the regressive systems tract, where precursor reefal facies such as *Cladocoropsis* and Stromatoporoid grainstones and rudstones are prevalent. Within these settings, dolomitization is intimately linked with the diagenetic process of replacement of precursor dissolution enhanced limestone, resulting in the formation of interconnected vug networks that significantly enhance fluid-flow pathways and impart favorable petrophysical properties to the reservoir. Detailed analysis of whole-core computed tomography (CT) scans reveals that the vuggy dolostones are characterized by elevated porosity and permeability values, highlighting the critical role of the precursor dissolution phase in creating high-quality reservoir rock. The observed correlation between dolomitization, dissolution, and the development of touching vug networks underscores the complex interplay between diagenetic processes and reservoir quality, with important implications for the accurate characterization and modeling of carbonate reservoirs. These findings suggest that the regressive systems tract, particularly areas with precursor reefal facies, may represent a key target for exploration and development efforts in these carbonate reservoirs, as this setting appears to be prone to the formation of high-porosity and high-permeability reservoir rocks.

Tight dolomite facies, interpreted as non-porous to poorly porous dolomitized intervals, are more commonly developed in the lower portions of the stratigraphic succession and display broad spatial continuity, predominantly across flank areas (Figure 9). Their lateral persistence and volumetric significance indicate a strong influence of stratigraphic architecture and precursor fine-grained depositional facies on dolomitization patterns and resultant reservoir quality. This tight dolomite facies is interpreted to result from pervasive burial related dolomitization processes. The mechanism is likely driven by compaction-induced fluid expulsion from deeper evaporitic formations which supplied magnesium-rich basinal brines. These fluids migrated vertically along stratigraphic discontinuities and laterally along facies boundaries, preferentially replacing limestone in mud-dominated, low energy facies of the transgressive systems tract sequence (Sun, 1995; Warren, 2000). Unlike porous dolomite, tight dolomite fabrics exhibit more uniform and laterally continuous geometries, indicative of early-stage diagenetic overprinting and limited preservation of primary porosity (Lucia, 1995). These observations are consistent with a model of thermally driven, regionally extensive dolomitization under burial conditions, with important implications for reservoir quality prediction and 3D property modeling in mature carbonate systems.

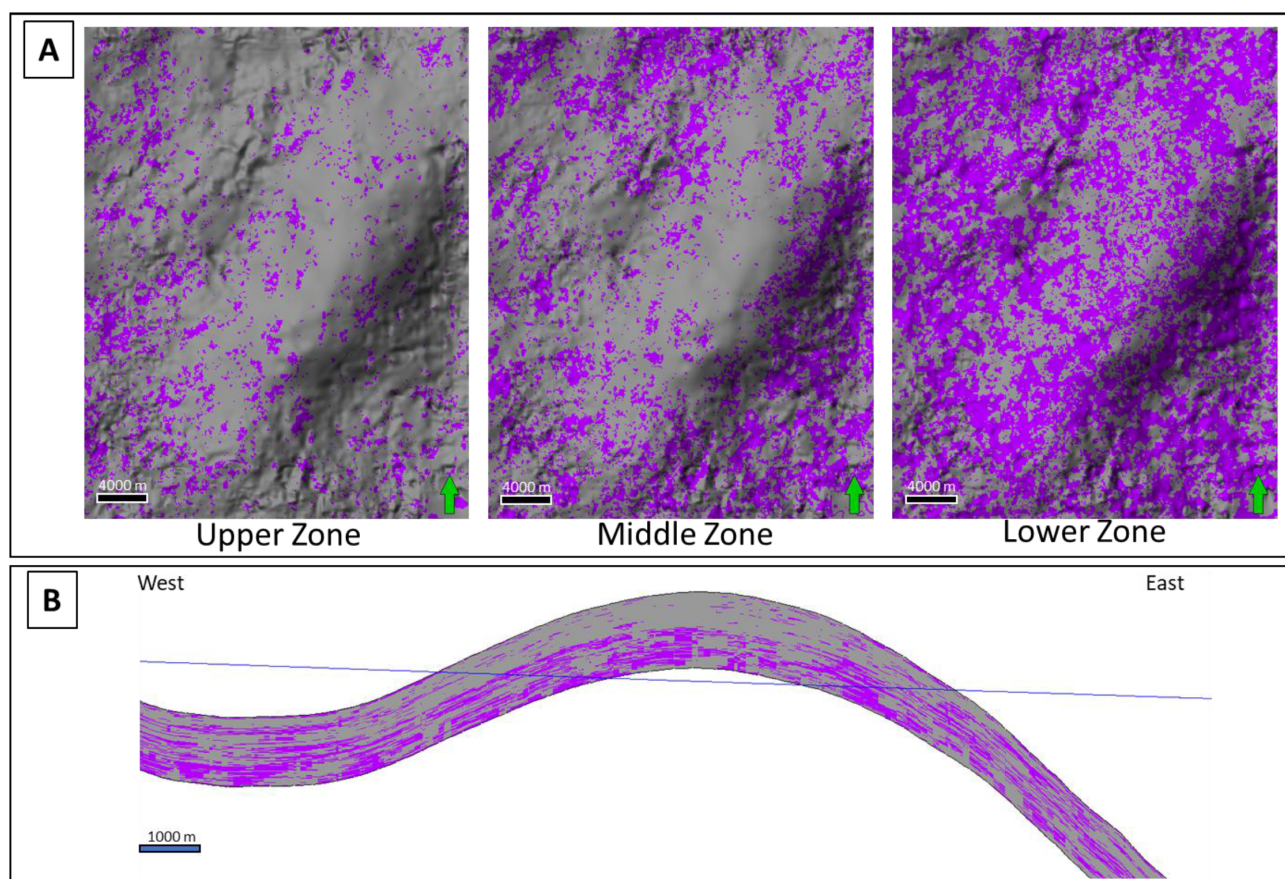


Figure 9—GHREM-based tight dolomite facies distribution results: (a). Map view of tight dolomite facies distribution in the upper, middle and lower zone; (b). Cross section view of the tight dolomite facies distribution

The GHREM workflow enabled the 3D prediction of the two distinctive dolomite geobodies with unprecedented resolution and accuracy, capturing a diverse range of geometries—variable in shape, size, and continuity. These geobodies represent the primary loci of porosity and permeability, making them fundamental to assessing and predicting reservoir quality. By leveraging data-driven pattern recognition, the GHREM approach effectively models the spatial complexity of dolomitization distributions and points to elements of their stratigraphic and structural controls. This predictive capability is critical not only for identifying sweet spots within the reservoir but also for guiding optimal field development strategies and

improving hydrocarbon recovery. These results highlight the essential role of integrated machine learning workflows that incorporate stratigraphic, structural, and diagenetic frameworks in reducing subsurface uncertainty and enhancing the predictive understanding of dolomite reservoir systems.

Conclusion

In conclusion, this study demonstrates the effectiveness of this machine learning-based workflow, termed GHREM, in predicting the 3D distribution of dolomite facies geobodies in a carbonate reservoir. By integrating high-resolution seismic attributes, stratigraphic zone properties, and well-log data, the GHREM approach achieves high predictive accuracy of ~85%, capturing the complex spatial relationships between porous and tight dolomite elements in the reservoir. The results reveal that porous dolomite is characterized by thin, laterally discontinuous streaks with limited spatial extent, primarily confined to the upper stratigraphic units and mid-flank to deep-flank structural positions, with isolated occurrences in crestal areas. In contrast, tight dolomite facies exhibit broad spatial continuity, particularly across flank areas, and are more commonly developed in the lower portions of the stratigraphic succession. The tight dolomites display more uniform and laterally continuous geometries, indicative of early stage diagenetic overprinting and limited preservation of primary porosity. The GHREM workflow provides a quantitative and objective method for modeling dolomite reservoirs, enabling the identification of sweet spots, optimization of field development strategies, and improvement of hydrocarbon recovery. The study highlights the importance of integrating machine learning with geological frameworks to reduce subsurface uncertainty and enhance predictive understanding of complex reservoir systems, ultimately contributing to more accurate reservoir characterization. By capturing the distinct distributions and geometries of porous and tight dolomite facies, this approach can inform targeted exploration and production strategies, maximizing the potential of these complex carbonate reservoirs.

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